

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Graphite / moderator slows the neutrons down (1) so they can be absorbed [by uranium atoms] (1) Boron / control rods absorb fission neutrons (1) so only 1 causes further fission / keeps the reaction at constant rate or converse (1)	4			4		
	(b)	(i)		2 (1) ${}_0^1\text{n}$ (1) Award 2 marks for ${}_0^1\text{n} + {}_0^1\text{n}$		2		2		
		(ii)		[The absorption of a neutron] makes the nucleus unstable (1) due to an imbalance between the numbers of neutrons and protons (1) Accept U-236 produced or increases the mass number	2			2		
	(c)	(i)		${}_{38}^{94}\text{Sr} \rightarrow {}_{39}^{94}\text{Y}$ (1) + ${}_{-1}^0\beta$ (1) [accept ${}_{-1}^0\text{e}$] Treat as neutral the next decay if shown	1	1		2		
		(ii)		It takes <u>75s</u> to halve the (1) {number of particles or atoms or nuclei / activity / mass / amount of the substance} (1) Treat as neutral reference to radioactivity	2			2		
		(iii)		[No] because half-lives for strontium and yttrium are short (1) so will decay into stable <u>zirconium</u> or a stable product [quickly] (1)	1	1		2		
				Question 6 total	10	4	0	14	0	0